EXECUTIVE SUMMARY

Banaras Hindu University is the largest and the first truly residential university of India. The confluence of oriental and theological learning with liberal arts, science and engineering, ayurveda and modern systems of medicine and agriculture makes BHU a unique Capital of Knowledge - '*Sarva Vidya ki Rajdhani'* in the words of its eminent past faculty Dr. S.S. Bhatnagar, FRS - where East meets West. The students passing out of the portals of such a Capital of Knowledge not only possess professional skills but are also imbued with a deep sense of Indian-ness.

The holistic model of education, conceived and enriched by its illustrious founder, Mahamana Pandit Madan Mohan Malaviyaji, offers unmatched perspectives to young minds and facilitates the accomplishment of their creative talents. Its contribution in extending the frontiers of knowledge in critical areas and also in the regeneration of community values is well manifested throughout the world with its alumni occupying key positions in varied professional domains. Its alumni and faculty members have been leaders in the national movement, in nation building after independence and in establishing major industries/infrastructure in as diverse sectors as steel,coal,minerals,energy,railways and water reservoirs/dams etc. Its alumni have been Vice-president of India, Chairman and Vice-chairman of Rajyasabha, Chief Ministers of states, Union Cabinet ministers, Vice-chancellors, Chairman of ONGC, SAIL, Coal India, Railway Board, Atomic Energy Regulatory Board etc, Directors of Indian Institute of Science, Indian Institute of Technology, Indian Institutes of Management, Director General of CSIR, DAE, DRDO Laboratories, Secretaries of Governemnt departments etc. They have been recipients of Bharatratna, Padma Vibhushan, Padmabhushan and Padmashree, Fellowships of Royal Society of London, Fellowships of Indian Academies of Science, Engineering, Medicine, Agriculture, Music, Literature, Dance and Drama etc. Its faculty members have been honoured with prominent awards and recognitions of the country and abroad such as Bhatnagar award (6), Fellowships of Indian National Science Academy (27), Rashtrapati Samman for Sanskrit scholars (6), B.C.Roy award and Jawaharlal Nehru fellowship and almost all well known fellowships and awards in the field of scholarship and research. University has maintained its prime position in the teaching and research and has recently been recognized as the **University with Potential for Excellence** (UPE) by the University Grants Commission (UGC) and the best University of India by the India-Today magazine.

It is one of the handful of Indian Universities with undergraduate and postgraduate education and research all in one campus, and of these the only one with a universal spread of disciplines.

The University is located on the western bank of river Ganga in the holy city of Varanasi. The picturesque main campus of BHU is spread over 1300 acres of land with majestic buildings of great architectural delight. It enshrines within its precincts a phenomenal range of Faculties incorporating diverse disciplines of Science, Humanities, Social Sciences, Commerce, Law, Education, Visual Arts, Performing Arts, Sanskrit Vidya Dharma and Vigyan, Management, Medicine – Modern, Ayurveda & Dental Science, Engineering and Technology, Agriculture, Library Science, Journalism and a large number of Indian and Foreign Languages. The University comprises **3 Institutes** (Institute of Medical Sciences, Institute of Agricultural Sciences and Institute of Environment & Sustainable Development), **16 faculties, 131 Departments, a Mahila Maha Vidyalaya** (Women's College) and 5 **interdisciplinary Schools**. The erstwhile Institute of Technology has been upgraded to **Indian Institute of Technology.** There are at present 18 departments which have received support under Special Assistance Programme (8 Centres of Advanced Studies and 10 Departments under DRS level), and 7 departments/ schools supported under FIST programme of DST.

It also has four colleges located in the city which are admitted to the privileges of BHU and three schools. In addition, the Rajiv Gandhi South Campus has been established in the year 2006 in a sprawling campus of 2600 acres located about 75 kms away from the main campus, in Barkachha, Mirzapur.

Research and Education

Research Priority Areas

The University has identified various research priority areas. It is expected that the respective faculties will direct their resources in these priority areas on a differentiated basis. There are special flagship areas with credible eminence such as : *Materials Science and Technology, Theoretical Condensed Matter Physics, Genetics and Biotechnology,Molecular Biology, Infectious Diseases, Environment & Ecology, Women Studies, Psychology, Conflict Resolution, Ayurveda, Indian Classical Music, Sanskrit and Dharm Vidya Vigyan.* These areas are to be nurtured to help them maintain international competitiveness. Some of the unique milestones of the University are:-

- BHU received **"University with Potential for Excellence"**status from UGC.
- BHU is the only University of India which has Indian Institute of Technology- I.I.T.(BHU) on its campus.
- Faculty of Veterinary Sciences has been approved by the Academic Council as well as the Executive Council to be established at R.G.S.C.
- Centre for Interdisciplinary Mathematical Sciences supported by DST.
- Centre for Genetic Disorders supported by DBT.
- Interdisciplinary School of Life Sciences supported by DBT.
- Establishment of Trauma Centre under PMSSY.
- Institute of Agricultural Sciences as Nodal Centre for Agricultural Innovation Partnership under USAID programme with several US Universities- Cornell, Georgia, Buffalo, UC Davis, Ohio, Tuskegee, Purdue and Illinois.
- Faculty of Dental Sciences.
- Strengthening & Development of Agricultural Education by ICAR, New Delhi.
- Experimental learning through the establishment of Bio-control lab, Tissue Culture lab, Fisheries and Hi-tech laboratories at Institute of Agricultural Sciences by ICAR.
- Sanction of Tandetron Accelerator facility to the Department of Physics.
- Strengthening of Space Science teaching and research in BHU by ISRO.
- Assistance for creation/seed infrastructure facilities at Institute of Agricultural Sciences, BHU by ICAR.
- Malaviya centre for Human Values and Ethics supported by Ministry of Culture.
- A unique Bhojpuri Adhyayan Kendra along with a Lok-Kala Sangrahalaya.

R & D Projects

The Vice-Chancellor has taken personal interest in encouraging the faculty members to generate additional resources for research activities through sponsored research projects. As a result of this initiative, there is a substantial change in the mindset and it is expected that more faculty members would have funding through sponsored research projects. The University has introduced several reforms in existing rules for smooth management of the research projects.

Industrial Collaborations

The improvements in the infrastructure of the University in general and the research laboratories in particular are expected to attract more industrial collaborations opening ways for additional funds. The University will provide all possible support for such collaborations.

Patenting and IPR

The University has already formed an IPR Cell to encourage the patenting culture. The University is already making provisions for meeting all the expenses related to filing of patents and it is proposed to enhance the budget of the IPR Cell further during XII-Plan.

Academic Reforms

Banaras Hindu University offers courses in the widest possible range of subjects. It is home to undergraduate, postgraduate and research programmes on the same campus, a unique feature for the Indian University System where most of the Universities are confined to postgraduate teaching and research only, leaving undergraduate teaching for colleges. Quality teaching at undergraduate and postgraduate levels requires a research environment. The Banaras Hindu University is unique in this respect also, as there is a strong tradition of research in practically all the disciplines. Capitalizing on these unique features, the University plans to embark upon a thorough restructuring of the existing degree programmes for making them more attractive and relevant to meet the challenges of the globalization. There is also a need for reviewing the existing admission and evaluation procedures.

Choice Based Credit System

The University has adopted semester system for all the degree courses and is presently following 'Choice Based Credit System' on a limited scale. it is planned to adopt a truly 'choice-based credit system (CBCS) and 'open sky' options allowing greater flexibility during the XII-Plan period. Thus for example, the students of fine arts can benefit tremendously by taking courses in computer graphics and ceramics. Similarly, engineering students can benefit from courses in Intellectual Property Rights, International Trade Law of the Faculty of Law. There are tremendous possibilities for such interdepartmental, inter-faculty interactions leading to a degree programme.

Entrance and Regular Examinations

University is utilizing internet facilities for the sale of forms, deposit of test fees issue of admit cards, conduct of tests, declaration of results and counseling of the eligible candidates for all entrance tests. The formalities of filling of forms, deposition of the examination fee and issuing admit cards are proposed to be done through the internet during the XII-Plan period.

Coordination with other Universities

The University believes that all the Central Universities should join hands in conducting a common entrance test for admissions to the various undergraduate, postgraduate and Ph.D.degree programmes, on the pattern of the Joint Entrance Examination conducted by the IITs. In due course of time, uniform credit-based evaluation system may be evolved enabling transfer of students with credit from one central university to another. The University proposes to take initiatives to form a joint forum of Central Universities to address such issues of common interest.

Internal –Quality assurance

Banaras Hindu University has been accredited by the National Assessment and Accreditation Council (NAAC) at 'A' level in the year 2006. The NAAC National Action Plan requires every accredited institution to establish an Internal Quality Assurance Cell (IQAC) as a post accreditation quality sustenance measure. Accordingly, IQAC has been setup in the University with the Vice-Chancellor as its Chairman. The IQAC is the central point in implementation of the Quality Policy of the university and work for quality enhancement and sustenance. It has developed a system for conscious, consistent and catalytic improvement in the performance of the university. IQAC will develop a quality manual describing various benchmarks and the processes designed to achieve them. The manual would describe the QA system, the set of guidelines, codes of good practices and procedures to be implemented by the various units.

Based on the prescribed guidelines, each unit of the university would develop its own internal quality assurance mechanism. The mechanisms shall be coherent with the quality assurance framework and approved by the IQAC, to evaluate the quality of teaching programmes and courses, academic staff, teaching and learning experience, student assessment, internal moderation, support services, resources and facilities, and research and programme review processes.

The Future Vision

In the coming years University shall strive to further promote excellence in quality of teaching and research. The road map for the next five years has been drawn with focus on expansion of research facilities, strengthening of teaching laboratories, expansion of student amenities, strengthening of the S.S. Hospital and making the Trauma Centre functional. Some of the highlights of new initiatives are enumerated below:

1. Cafeteria-cum- Guest House Complex

The University has grown tremendously in size and reach. In order to ensure visibility on national and international platforms, it is essential to have world class infrastructure for exchange of ideas. Through this, multidisciplinary scientific interactions may take place among the students and between teacher and students. University is making all possible efforts to make our students capable of coping with emerging challenges in the era of global completion and rapidly changing higher education demands. Keeping this in sight, it is proposed to build a 12 storeyed **Cafeteria-cum- Guest House Complex** with an area of 1000 sq m per floor. The basement, ground and first floor of the building shall be used for parking facilities and three floors shall be allotted for Cafeteria. The cafeteria shall have a catering capacity of ~1000 persons per floor. Four floors shall be utilized for Guest House (200 rooms). The remaining two floors shall be used for other community activities.

2. International Hostel

In order to attract students and academicians from all corners of the world it is essential to provide them all facilities for a comfortable stay in the campus. The available infrastructure is not sufficient for accommodating increasing number of foreign students. It is therefore proposed to construct an **International Hostel** with 500 rooms for students. Each room shall have a kitchenette and attached bathroom facility. There will also be 40 suites for visiting foreign faculty (studio apartments). The hostel shall have a Launderette, Lounge, Committee room, Indoor sports and recreation facilities. Each room in the hostel shall be equipped with Internet facility.

3. Centralized Instruments Facility

The University plans to establish a centralized facility for housing major research equipment. The facility shall have state-of-the-art equipment for research in basic and applied sciences. This facility will have dedicated technical staff for operation and maintenance of equipment. The technical staff will also take care of sophisticated equipment in various departments/schools. By establishing this facility University desires to provide a centralized work space having state-of-the-art facilities and equipment. The Centralized Instrumentation Facility will be in addition to the existing central instrumentation laboratories in the departments and faculties and it will improve the scientific environment and bring about outputs of international standards.

4. Core Computational and ICT Facility

It is proposed to establish a core computational facility which shall cater to intensive computing, requirements of the faculty members, research scholars and other students. The facility shall have high performance computing facility for simulation, modelling and data analysis. It will also cater to the ICT requirements of the University. It is proposed to enhance the internet band-width capacity to 20 GB.

5. New Multi-storeyed Mega Hostel Blocks

University proposes to construct four **Multi-storeyed Mega Hostel Blocks** with 1000 rooms in each in order to increase hostel accommodation capacity to accommodate 60% students as against 45% at present. It is proposed to add 1000 single occupancy rooms Ph.D. scholars during the XII-Plan. Further, with the increase in number of girl students, it is proposed to earmark 1000 rooms for girl students. 2000 rooms shall be available for boys. These Hostel blocks shall provide accommodation to students of all the faculties. Every hostel is to be equipped with modern kitchen and launderette, in-door sports and recreational facilities and internet connectivity in every room.

6. Building for Languages Departments

It is proposed to construct a multi-storeyed building for housing the language departments at one place. It will reduce congestion in the old Arts Faculty building. It will also have a well-equipped Media lab with modern equipment and studio. The Media lab would be provided with a mobile Audio Visual Recording Van. It is proposed to set-up a language lab also in the same building.

7. Residential Quarters

In order to attract and retain talented scholars as faculty members it is necessary to provide them with appropriate accommodation on the campus. Keeping in view the shrinking space and greenery it has been decided by the University to adopt the policy of vertical expansion in all its new buildings. In this spirit it is proposed to construct a multi-storeyed, walled- complex consisting of several blocks of residential accommodation of different categories. This complex shall have 400 flats for teaching faculty and 250 flats for other employees and adequate parking space on the ground floor. The complex will be equipped with a central security system, recreational area and children's park.

8. Strengthening of Hospital Services

Sir Sunderlal hospital is the teaching and training hospital of Institute of Medical Sciences. It caters to a large population of eastern region and there has been substantial increase in the patients and clinical load during last few years. At present, this is the single tertiary care hospital in the region providing speciality and superspeciality services to nearly 20 crores population of the Eastern UP, Western Bihar, adjoining Madhya Pradesh, Chhattisgarh and Jharkhand states as well as neighboring county of Nepal. Most of these patients belong to poor socio economic status; therefore, providing healthcare at very low cost is a big challenge. It is proposed to strengthen the hospital infrastructure by providing additional man power and equipment. A Robotic Surgery Theatre with remote control option and Gamma Knife Unit for treating brain lesions (a non invasive technique) is also proposed for providing the students and faculty members and opportunity to acquire these skills to treat patients more effectively.

In the UPE scheme, BHU has been sanctioned money to establish a Stem Cell Research and Bone Marrow Transplant Facility at **Sir Sunder Lal Hospital.** University has now established a ***Centre for Stem Cell and Bone Marrow Transplant*** which will help cases of heart ailment, muscle injuries, spinal cord injuries, cancer treatment, and damaged liver regeneration. Under the *Pradhan Mantri Swasthya Suraksha Yojana (PMSSY)*, infrastructure has been established for a **Trauma Centre** having multidisciplinary departments of Orthopedics, Neurosurgery, Cardiology, Burn Unit, and many others. This will go a long way in saving lives of the unfortunate trauma victims in the region by an efficient and coordinated management. This centre should become functional as early as possible.

BHU also proposes to create a new Super-Specialty Hospital for furthering the capabilities of Sir Sunder Lal Hospital to serve the people of this region.

9. A GLP Compliant Animal House with Transgenic Gene Knockout Facility is proposed to be created for advanced research in life sciences.

10. BHU also aims to establish an innovative **Institute of Tribal and Genomic Medicine** in the Rajiv Gandhi South Campus (RGSC) of Banaras Hindu University. The establishment of the institute will not only give fillip to the research and development of quality traditional medicines existing in the tribal belt, but also in finding the molecular basis of these medicines for their global acceptability. This centre shall be established by garnering financial support of appropriate ministries/ departments.

11. The Rajiv Gandhi South Campus (RGSC) of Banaras Hindu University is all set to get a new look with the start of XII plan. Steps have already been started to establish a **Faculty of Veterinary Sciences** at RGSC. This faculty shall provide contemporary education & training to meet the changing needs of the veterinary & animal sciences for better animal health, production and processing. The faculty shall also provide referral & consultancy services and efficient extension services to the stake holders in all aspects of veterinary and animal science. It will produce competent, skilled, socially relevant professionals as the human resource for conducting basic & innovative research in veterinary & animal sciences. The faculty shall provide continuous education to livestock farmers by dissemination of technologies for animal health and production.

12. The University proposes to establish an **Institute of Translational Research** for application of knowledge of modern biology into clinical care. It will help in systematic collection and analysis of large amounts of clinical data, development of ways and means of personalized medicine, development of specific stem cell populations and genetic diagnosis carrier analysis, prenatal and predictive diagnosis and counselling.

13. Infectious diseases are one of the serious concerns in the region and taking a note of this BHU is proposing to establish a **"Biological Containment Level Four Facility"** at RGSC for research in such areas.

14. Diabetes, Coronary Artery Diseases, Cancer, Renal and Gastrointestinal Diseases are on the rise and their treatment is highly cost intensive. In order to provide medical facilities for these diseases at an affordable cost it is proposed to establish a **Superspeciality Hospital** on a Hub & Spoke Model wherein the central facilities like Radio diagnosis and Pathology, which are required by all specialities will form the hub and all superspecialities will form spokes around this hub. This hospital shall have Institute of Cardiac Sciences, Centre for Endocrine Disorders, Centre for Kidney Diseases and Transplant, Centre for Neurological Science & Mental Health, Centre for Gastro Intestinal and Hepatic Diseases and a Comprehensive Cancer Centre.

15. Nano and Micro fabrication Laboratory

The Nano Micro Fabrication lab would consist of three large areas indicated by their cleanliness level. The class 10,000 area will house chemical mechanical polishing, wafer thinning, packaging and assembly, and MBE. The class 1000 and 100 areas will house the rest of the processing capabilities, including metal deposition, lithography, oxidation, dielectric deposition, plasma processing, and wet-chemical processing. The major fabrication facilities include thin film deposition, plasma processing, optical and electron beam lithography, thermal processing, electroplating, wafer lapping, polishing, dicing, bonding and sawing, wire bonding, flip chip bonding, MBE growth, a 2-metal/2-poly CMOS line and a MEMS line. Characterization tools include optical and electron microscopy, X-Ray tomography, grazing angle diffraction and reflectivity, mass-spectroscopy, AFM, PFM, MFM, STM, stylus and optical profilometer, ellipsometer, low/high force scanning nano-indentation tribology, surface analysis tools, and high speed electronic and optical testing. These facilities would be open to the multitude of user community from a wide range of disciplines including materials, electronics, mechanical, chemical, and biomedical engineering as well as physics, chemistry, and biology. Its research will include: nanostructured ferroics and multiferroics, high k materials, MEMS, and Si electronics, SiC electronics and molecular electronics.

16. Convention Centre:

The University proposes to augment its infrastructure for hosting world-class events, whether they are conventions for 10,000 delegates, seminars for 1000 or meetings for just 250 persons. For this it has been planned to build a **Convention Centre** having state-of-the-art infrastructure, design, technology, telecommunications and AV equipment. The main hall of this centre will be a Pillar-free internal hall that can hold a 10,000-delegate plenary and can be portioned into 6 smaller halls for holding simultaneous sessions of smaller groups. The Convention Centre shall have a spacious pre-function lobby area for organizing exhibition, poster presentation and reception. It will also have 10 breakout rooms, 10 specialized seminar halls, 5 boardrooms and a VIP lounge. The design of this Convention Centre shall meet International standards of engineering and environment regulations. It will have sufficient space for parking. It is proposed to create at least 50 suites for accommodating speakers and dignitaries. An in-house banqueting service shall be provided. The Convention centre shall be a unique facility for promoting academic interactions on a large scale.

17. Expansion of Bharat Kala Bhawan :

Bharat Kala Bhavan (BKB) is an art and archaeological museum. BKB museum of BHU is the unique one among Indian Universities having a collection of about 1.10 lacs and as per international norms a small part of this collection has been displayed in 13 galleries. Several students, research scholars and teachers from all over the country and abroad visit this museum for their higher education and research purposes. It is proposed to give this museum a new look and better infrastructure during XII-Plan. The University plans to open a Science Gallery for displaying scientific artefacts, old medical equipment and tools for exhibiting advancement in science during past decades. It is also proposed to create a seminar hall with modern facilities.